HOMEWORK 2.3 · UNIT 2 · NC.8.EE.1

# Power-to-a-Power; Negative & Zero Exponents

Targets: EE.1-2, EE.1-3, EE.1-7, EE.1-6, EE.1-4, EE.1-8, EE.1-9

Name _____ Date _____ Block _____

**Remember: Power to a power → multiply**: $(x^2)^3 = x^6$. Raise every factor: $(3x)^2 = 9x^2$. **Zero:** any nonzero base$^0 = 1$.
**Negative:** flip to a fraction: $2^{-3} = \frac{1}{2^3} = \frac{1}{8}$, $5x^{-2} = \frac{5}{x^2}$.

**SCORE 3.0** Approaching · the basics

Targets: EE.1-2, EE.1-3

**1** Simplify. Write as one power. $(5^2)^3$ EE.1-2

**2** Simplify. $(y^5)^4$ EE.1-2

**3** Find the value. $4^{-2}$ EE.1-3

**4** What is the value of $(-6)^0$? EE.1-3

Ⓐ −6 Ⓑ 0 Ⓒ 1 Ⓓ −1

**5** Simplify. $\frac{x^9}{x^2}$ REVIEW 2.2 EE.1-2

**SCORE 4.0** Proficient · on target

Targets: EE.1-7, EE.1-6, EE.1-4

**6** **Lou** made a mistake. Circle it. Explain it. Fix it. EE.1-7

LOU'S WORK
$(x^3)^4 = x^7$

**7** Simplify. Use positive exponents. $(2x^{-3})^2$ EE.1-6

*Hint: Square the 2 too.*

**8** Simplify. $(2a^4)(7a^3)$ REVIEW 2.2 EE.1-4

**SCORE 5.0** Mastery · go deeper

Targets: EE.1-8, EE.1-9

**9** Use the pattern. Explain why $2^{-2} = \frac{1}{4}$. EE.1-8

| Power | $2^3$ | $2^2$ | $2^1$ | $2^0$ | $2^{-1}$ | $2^{-2}$ |
|---|---|---|---|---|---|---|
| Value | 8 | 4 | 2 | ? | ? | ? |

Each time the exponent goes down 1, you _____. So $2^{-1}$ = _____ and $2^{-2}$ = _____.

**10** Kim says $3x^{-2} = \frac{1}{3x^2}$. Do you agree? Explain. EE.1-9

I _____ because the exponent is only on _____.